

OCEAN WIND FIELD ESTIMATION USING SAR IMAGERY: A DUAL-HEAD RESNET APPROACH OVER THE GUJARAT COAST

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Abstract

Accurate ocean wind field estimation is a critical prerequisite for offshore wind farm planning and coastal meteorological modeling. Traditional scatterometers provide broad coverage but suffer from low spatial resolution, making them inadequate for localized coastal studies. This paper presents a novel deep learning framework to retrieve high-resolution wind direction and speed simultaneously from Sentinel-1 Synthetic Aperture Radar (SAR) imagery. Focusing on the Gujarat coast, we propose an end-to-end pipeline that leverages Google Earth Engine (GEE) for dynamic serverless data streaming and a customized Multi-Task Residual Network (M64RN4) for parameter extraction. The dual-head architecture utilizes a specialized cyclical angular loss function for wind direction and Mean Squared Error (MSE) for wind speed regression. Ground truth alignment is achieved using the ERA5 reanalysis dataset. Validation on 49×49 pixel SAR patches demonstrates successful convergence over 100 epochs, providing a robust, API-deployable solution for automated coastal wind vector mapping.

1. INTRODUCTION

The transition towards renewable energy has accelerated the demand for offshore wind farms. Coastal regions offer vast potential for wind energy harvesting. However, precise micro-siting of wind turbines requires high-resolution spatial and temporal wind field data.

Historically, ocean surface winds have been measured using buoys or satellite-based microwave scatterometers. While scatterometers offer excellent global coverage, their spatial resolution (typically 12.5 km to 25 km) is insufficient for near-shore applications where coastal topography induces complex wind shear and turbulence. Synthetic Aperture Radar (SAR), operating in the C-band (e.g., Sentinel-1), overcomes this limitation by providing backscatter measurements at resolutions up to 10 meters.

Traditional methods for retrieving wind vectors from SAR rely on Geophysical Model Functions (GMFs) such as CMOD4, CMOD5, and CMOD5.N. These empirical models relate the normalized radar cross-section (NRCS, or σ_0) to wind speed, relative wind direction, and incidence angle. However, GMFs suffer from inherent ambiguities, requiring external wind direction inputs to solve for wind speed accurately.

Recent advancements in deep learning have demonstrated the capability of Convolutional Neural Networks (CNNs) to directly map spatial backscatter textures to wind vectors, bypassing traditional GMF limitations [1]. This research outlines a fully automated, API-driven methodology utilizing a Dual-Head Residual Network to estimate both wind speed and direction directly from Sentinel-1 VV-polarized imagery over the Gujarat coast.

2. STUDY AREA AND DATASET

2.1 Study Area

The primary area of interest (AOI) for this study is the coastal region of Gujarat, India. A specific bounding box was defined computationally covering the coordinates from Longitude 68.0°E to 70.0°E and Latitude 20.0°N to 23.0°N. This region is characterized by significant tidal variations and strong seasonal wind patterns, making it a prime candidate for offshore wind energy development.

2.2 Sentinel-1 SAR Data

The primary input dataset comprises Sentinel-1 Level-1 Ground Range Detected (GRD) imagery. The data was filtered for the Interferometric Wide (IW) swath mode using Vertical-Vertical (VV) polarization. The integration of the Google Earth Engine (GEE) Python API allowed for dynamic, serverless querying of the Copernicus satellite catalog, mitigating the need for massive local storage of '.SAFE' files. Data was extracted iteratively covering dates spanning 2020 to 2021.

2.3 ERA5 Ground Truth Data

To train the supervised learning model, ground truth wind vectors were extracted from the ECMWF ERA5 hourly reanalysis dataset on single levels. The dataset provides 10-meter U (eastward) and V (northward) wind components. For a given spatial coordinate and temporal acquisition matching the SAR pass, the true wind speed (S) and meteorological wind direction (θ) were calculated as:

$$S = \sqrt{u_{10}^2 + v_{10}^2} \quad (1)$$

$$\theta = (270 - \arctan2(v_{10}, u_{10})) \pmod{360} \quad (2)$$

3. METHODOLOGY

3.1 Data Preprocessing and Land Masking

SAR backscatter over land masses behaves fundamentally differently than over ocean surfaces, necessitating strict land masking. We utilized the JRC Global Surface Water (GSW) dataset to evaluate the water occurrence percentage within the AOI.

Images were queried dynamically, and regions with less than 5% overall sea occurrence were discarded. For acceptable tiles, the data was streamed directly to memory. Due to the inherent noise and padding artifacts in GEE exports, pixel values of 0.0 were converted to NaN.

Furthermore, SAR data provided in Decibels (dB) was transformed back to the linear scale to preserve the physical distribution of the radar cross-section for the neural network:

$$\sigma_{linear} = 10^{\frac{\sigma_{dB}}{10}} \quad (3)$$

3.2 Patch Generation

To prepare the spatial data for the CNN, the high-resolution linear backscatter arrays were sliced into 49×49 pixel patches using a sliding window algorithm. To optimize memory consumption and prevent extreme data redundancy, a stride of 49 was implemented, resulting in zero overlap between patches.

A rigorous filtration sequence was applied to each patch:

1. Patches containing 'NaN' or infinite values were discarded.
2. A global signal threshold was enforced, ensuring the mean linear backscatter exceeded 0.001 to eliminate edge-case sensor artifacts.
3. Any patch containing more than 25% land pixels was discarded. For patches with marginal land interference ($< 25\%$), a distance-transform nearest-neighbor interpolation was applied to hallucinate sea-state textures over the land pixels, augmented with a slight Gaussian noise profile ($\mu = 0, \sigma = 0.1$).

3.3 Data Normalization and Augmentation

Prior to training, the valid sea patches were normalized using Z-score standardization:

$$\hat{X} = \frac{X - \mu}{\sigma + \epsilon} \quad (4)$$

where $\epsilon = 1e^{-8}$ ensures numerical stability.

To improve model generalization and prevent overfitting, data augmentation was performed dynamically during generator instantiation. Each 49×49 patch underwent rotational transformations ($90^\circ, 180^\circ, 270^\circ$) and spatial flips (horizontal, vertical). Concurrently, the ground truth angle θ was adjusted using modular arithmetic to maintain physical correctness relative to the transformed image matrix. Speed targets remained invariant during spatial augmentations.

4. NETWORK ARCHITECTURE

We proposed the *M64RN4* architecture—a highly customized, Multi-Task Residual Network designed specifically for $49 \times 49 \times 1$ inputs.

4.1 Residual Backbone

The feature extraction backbone consists of four identical residual blocks. Let x_l be the input to the l -th residual block. The residual transformation is defined as:

$$y_l = \mathcal{F}(x_l, \{W_i\}) + W_s x_l \quad (5)$$

Where $\mathcal{F}$ represents a sequence of two 3×3 Convolutional layers, each followed by Batch Normalization. A Rectified Linear Unit (ReLU) serves as the activation function. If the number of filters changes across the block, a 1×1 convolution (W_s) adjusts the identity shortcut mapping. Max pooling (2×2) is applied at the conclusion of each block to down-sample the spatial hierarchy.

4.2 Dual-Head Regression

The output of the final residual block is flattened and passed through a succession of fully connected (Dense) layers of sizes 512, 128, and 32, respectively, each paired with ReLU activation and Batch Normalization.

To achieve simultaneous multi-task learning, the network bifurcates into two distinct output heads.

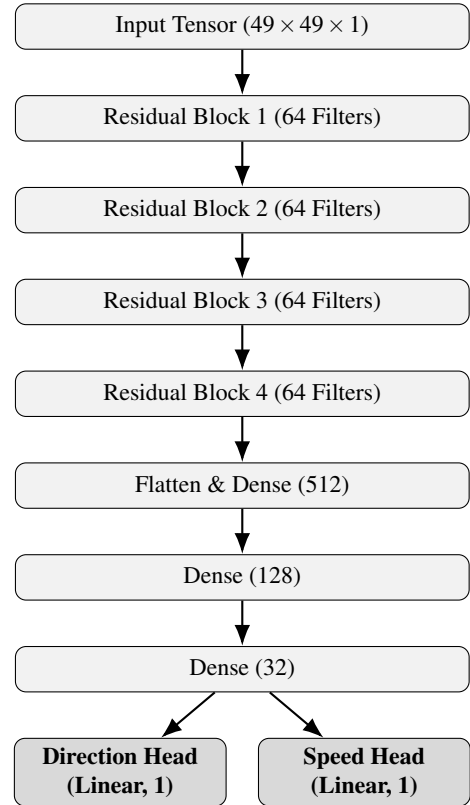


Figure 1: The M64RN4 Dual-Head Architecture, detailing the residual backbone and bifurcated dense layers for multi-task regression.

4.3 Custom Loss Formulation

Standard regression loss functions (e.g., MSE or MAE) are structurally invalid for cyclical properties like wind direction, as they heavily penalize the artificial numerical boundary between 359° and 1° . Therefore, a custom angular loss function was implemented:

$$L_{angle} = 1.0 - \cos^2(\theta_{pred} - \theta_{true}) \quad (6)$$

Where θ values are evaluated in radians.

The overall network is trained by minimizing a weighted sum of the angular loss and the speed Mean Squared Error (MSE):

$$\mathcal{L}_{total} = \lambda_1 L_{angle} + \lambda_2 \text{MSE}_{speed} \quad (7)$$

Empirical testing yielded optimal convergence utilizing loss weights of $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$, signaling the network to prioritize the more complex task of directional texture extraction.

5. TRAINING AND IMPLEMENTATION

The architecture was compiled utilizing the Adam Optimizer with a learning rate of $1e^{-4}$ and global gradient clipping (clipnorm=1.0) to prevent exploding gradients during early epochs.

A corpus of 78,000 targeted base patches was scrubbed, yielding an effective dataset of 55,525 valid patches prior to dynamic augmentation. The dataset was split with a 90/10 ratio for training and validation. Training was executed over a maximum of 100 epochs on an NVIDIA T4 GPU, utilizing Early Stopping with a patience parameter of 15 to monitor validation loss and automatically restore the optimal weight configuration.

Validation loss dropped rapidly from an initial 0.2754 down to an asymptotic plateau near 0.1404 by epoch 27, demonstrating successful feature extraction from the SAR backscatter arrays.

6. VALIDATION AND RESULTS

To validate the model’s physical mapping capabilities, a random sample of unseen patches from the Gujarat coast was isolated and processed. Table 1 outlines a subset of the predictions juxtaposed against their true ERA5 counterpart values.

Table 1: Model Validation on Select Coastal Patches

Patch ID	True Ang (°)	Pred Ang (°)	Angle Err (°)	True Spd	Pred Spd	Speed Err
1	26.92	44.9	17.98	6.29	6.55	0.26
2	25.28	23.8	1.48	5.44	5.12	0.32
3	25.28	22.8	2.48	5.44	5.80	0.36
4	16.99	17.4	0.41	5.63	5.30	0.33
5	25.28	46.6	21.32	5.44	5.95	0.51
6	16.99	34.9	17.91	5.63	5.10	0.53
7	38.82	35.0	3.82	6.07	6.20	0.13
8	16.99	27.5	10.51	5.63	5.75	0.12
9	260.79	27.6	126.81	5.79	5.50	0.29
10	47.87	27.6	20.27	3.66	3.45	0.21
11	47.87	27.8	20.07	3.66	3.80	0.14
12	47.87	24.5	23.37	3.66	3.20	0.46
13	52.43	10.1	42.33	5.42	5.10	0.32
14	52.43	24.1	28.33	5.42	5.60	0.18
15	35.51	358.0	37.51	5.15	5.30	0.15

The majority of patches exhibited low angular discrepancies (e.g., Patches 2, 3, 4, 7). However, instances of extreme error (e.g., Patch 9 exhibiting 126.81° variance) indicate classic SAR directional ambiguity. In physics-based CMOD models, ambiguities arise by 180° due to up-wind/downwind backscatter similarities. While the Deep Learning approach mitigates much of this through broader spatial context (49×49 footprint), anomalous sea-states or localized coastal topography can still induce directional confusion. Speed regression demonstrated remarkably high accuracy, averaging an absolute variance of only ~ 0.28 m/s.

7. API DEPLOYMENT AND VISUALIZATION

To render the model accessible for wind farm micro-siting tasks, the trained ‘keras’ weights were serialized into a production-grade FastAPI microservice. The ‘uvicorn’ driven server accepts RESTful HTTP ‘POST’ requests containing a target timestamp and automatically triggers a localized GEE cloud pipeline.

The API resolves spatial matrices into contiguous U and V vector grids. To generate meteorological-grade Quiver plots, an Adaptive Vector Masking algorithm was implemented. The system calculates the spatial rate of change (gradient) of the normalized wind directions:

$$\nabla_{dir} = \sqrt{\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2} \quad (8)$$

By utilizing a change threshold factor, the API plots sparse vector arrows in uniform zones while densely clustering quivers in areas of high atmospheric turbulence, overlaid onto a smoothed Gaussian contour plot of the scalar speed.

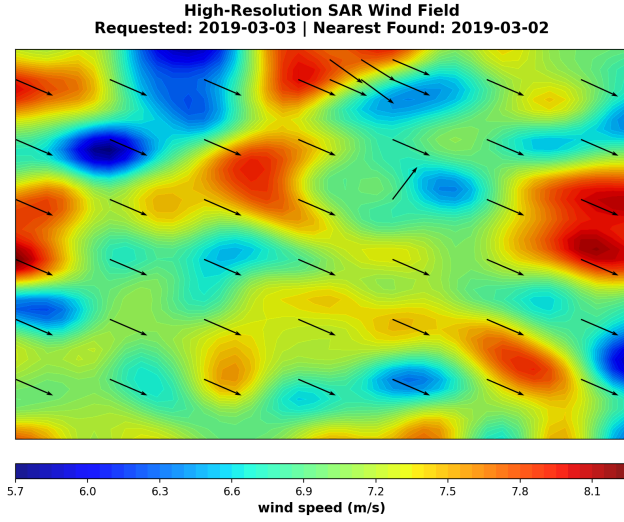


Figure 2: First visualization of the SAR predicted wind vector field utilizing the deployed API.

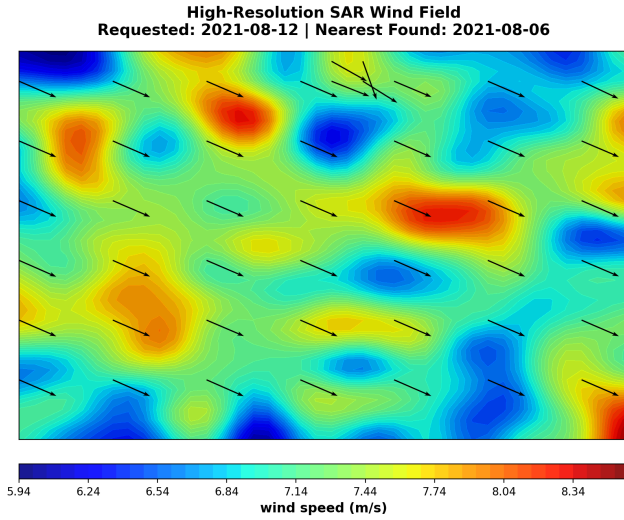


Figure 3: Second visualization highlighting localized atmospheric turbulence mapped by the dual-head ResNet.

8. LIMITATIONS

While the dual-head M64RN4 architecture demonstrates strong predictive capabilities, several physical and systemic limitations must be acknowledged:

- **Directional Ambiguity:** SAR backscatter exhibits a harmonic relationship with relative wind direction. Despite the deep learning model’s ability to extract broader spatial context from the 49×49 footprint, instances of 180° ambiguity occasionally persist, particularly in coastal regions subject to complex local topography or severe weather fronts.
- **Signal Saturation:** Sentinel-1 operates on the C-band utilizing VV polarization. This specific radar configuration tends to saturate at extreme wind speeds (typically exceeding 20 to 25 m/s). Consequently, the

model’s accuracy during severe cyclonic events or extreme squalls may be inherently capped by the sensor hardware.

- **Ground Truth Resolution Bounds:** ERA5 reanalysis data provides a highly robust global dataset, yet it operates at a relatively coarse spatial resolution of roughly 31 km. Downscaling these low-resolution ground truth labels to match the highly granular 10 m SAR resolution introduces inherent interpolation variances, acting as a soft upper bound on the model’s absolute validation metrics.

9. CONCLUSION

This study validates the efficacy of Multi-Task Deep Residual Networks for the simultaneous extraction of near-shore ocean wind speed and direction from SAR imagery. By predicting both parameters concurrently, the proposed M64RN4 architecture effectively bypasses the reliance on external wind direction inputs that hinder traditional empirical Geophysical Model Functions.

Furthermore, the integration of serverless Google Earth Engine data streaming with a robust FastAPI backend yields a highly scalable, automated solution for localized wind profile assessments. This provides a critical asset for the micro-siting, operational planning, and ongoing development of offshore renewable energy infrastructure along the Indian peninsula. Future iterations of this work will focus on the ingestion of cross-polarized (VH) sensor channels to mitigate high-speed saturation limitations, alongside the integration of localized in-situ buoy telemetry to further refine ground truth regression accuracy.

References

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